
A Validated Black-Box Instrument for Language-Model Dynamics: Reproducing Known Metrics with a Token-Lattice Cellular Automaton

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Abstract

1 We turn a language model into a *cellular automaton (CA)* over *token space*—
2 a ring of token cells updated in place by the model’s own windowed condi-
3 tional $p_r(x_i \mid x_{i\pm r})$, with common-random-number (CRN) damage spreading
4 as the probe—and develop it into a *validated* black-box measurement instrument.
5 The central move is **validation by reproduction**: before measuring a language
6 model, the instrument recovers established dynamical metrics on systems where
7 they are known. Its CRN damage-spreading estimator separates elementary CA
8 rules whose damage survives from those whose damage dies (ignition probability,
9 Cohen’s $d=3.0$); reproduces a stochastic discrete PCA’s damage spreading **bit-**
10 **exactly**—on the line where that model’s damage field is provably the automaton
11 itself, an independent prediction matches every cell over 4096×1500 updates
12 with zero mismatches, while an off-line control mismatches as it must—giving
13 the apparatus a calibration with no error bar attached; and—through its attrac-
14 tor census—recovers the known transition matrices of synthetic Markov sources
15 (self-recovery total-variation 0.22 vs. cross 0.95). We are explicit about what
16 the smooth-limit rungs do and do not establish: a renormalized tangent estima-
17 tor reproduces the logistic map’s analytic Lyapunov exponent to $O(\epsilon)$, but that
18 agreement is a property of the infinitesimal limit, and a token flip is $O(1)$ in a dis-
19 crete alphabet—so we keep that rung as a *unit test of the growth-rate arithmetic*,
20 not as validation of the instrument. Every measurement is certified against ap-
21 paratus confounds by an exact-zero-null CRN protocol and attributed at the level
22 of *statistics*, not *trajectories*. Applied to trained LMs, the validated instrument
23 yields black-box criticality measurements not otherwise available from weights-
24 free access: a **token-space Lyapunov exponent** and a diversity-controlled **damp-**
25 **ing length** of the token-lattice construction, an **effective interaction radius** at
26 which the model builds structure ($r \approx 4$), a model-invariant **damage light cone**,
27 and an **attractor census** of internalized structure. Finally we map a *boundary*: a
28 controlled cross-level study (two model families, two within-model axes) shows
29 the black-box token-space criticality does *not* proxy white-box activation-space
30 criticality—they are distinct, non-proxying levels, because the activation measure
31 is architectural. The contribution is a **validated instrument and the quantities**
32 **it measures**; the phenomena it lands on—criticality, computation at the edge of
33 chaos [Langton, 1990, Bertschinger and Natschläger, 2004, Zhang et al., 2025]—
34 are decades old, and we measure them from the outside rather than claim to dis-
35 cover them.

1 Introduction

Language models are evaluated almost entirely as functions: given a context, how good is the next-token distribution? But a masked or autoregressive conditional also *defines a dynamics*—iterate it in place over a lattice of tokens and one obtains a stochastic cellular automaton whose rule is the model. Iterated-map views of LLMs have appeared qualitatively [Utimula, 2026, Perez et al., 2024, Wang et al., 2025] and as mixing-time studies of masked-LM Glauber dynamics [Sana et al., 2026]; the phenomena a CA vocabulary names—damage spreading, light cones, Lyapunov exponents, attractor censuses [Hanson and Crutchfield, 1997]—are decades old. What has been missing is not a new phenomenon but a *measurement instrument whose readings can be trusted*: a small, controlled apparatus, calibrated against ground truth, with apparatus confounds certified away.

This paper’s organizing principle is **validation by reproduction**. We build the instrument—a token-lattice CA with CRN damage spreading (§2)—and, before turning it on a language model, show that it recovers *known* dynamical metrics on systems where the answer is established (§3): a decisive survives-vs-dies criticality split among elementary CA rules, a stochastic PCA’s published damage-spreading boundary (bit-exactly, where an exact mapping exists), the known transition matrices of synthetic Markov sources, and—as explicitly labelled smooth-limit arithmetic checks—the logistic map and a coupled-map lattice. An instrument that reproduces exact values on known systems can be *trusted* on an unknown one. Only then do we report the new, weights-free measurements it yields on trained LMs (§4): a token-space Lyapunov exponent, a damping length, an effective interaction radius, a damage light cone, and an attractor census. Finally we map the instrument’s *boundary*—the cross-level regime where the black-box token-space reading provably decouples from white-box internals (§5). The edge-of-chaos and criticality ideas the results touch are old [Langton, 1990, Bertschinger and Natschläger, 2004, Zhang et al., 2025]; our contribution is the validated black-box *measurement* of them, not their discovery.

Simplicity and tested reliability. A deliberate choice runs through the paper: we use only *simple, mature* primitives from cellular-automaton and dynamical-systems theory—single-site damage spreading, the damage-spreading Lyapunov exponent [Bagnoli et al., 1992], the light cone [Lieb and Robinson, 1972], the attractor census [Hanson and Crutchfield, 1997]—rather than bespoke machinery. Each has been used and tested for decades, and the value of transferring them is precisely that they are *understood*: their expected behaviour on known systems is a fixed benchmark. That is what the validation ladder exploits (§3)—reliability here is not asserted, it is *demonstrated* by reproducing exact known values. The stance of the paper is a disciplined transfer of a reliable field’s instruments to language models, with every reading either calibrated against ground truth or certified against apparatus confounds, and with the one place the transfer fails reported as plainly as the places it succeeds.

2 The instrument

Rule. A window of $w = 2r + 1$ tokens with the center masked is fed to a masked LM; the tempered center distribution is the kernel $p_r(x_i \mid x_{i \pm r})$. For an autoregressive model (§4.5) the window is the r cells to the left and the rule is the next-token distribution—an order- r Markov approximation, the causal analog of the symmetric masked window. **Automaton.** N ring cells, async random-order single-site Glauber (synchronous updates manufacture period-2 blinkers and are only an apparatus arm [Wang et al., 2025]); special tokens forbidden as emissions. **CRN coupling.** Twin lattices sharing initial state, update order, and the uniform stream driving inverse-CDF sampling differ only in the factor under test; the *null* arm must diverge by exactly zero (it does, on both masked and autoregressive paths), certifying the coupling. **Four pillars:** $T \times r$ phase sweeps; damage spreading (block flips, CRN twins, ignition probability separate from spread); attractor census (random soup to quasi-stationarity, validated against a corpus); and differential certification—attribute at the level of *statistics*, never trajectories (under chaos, trajectory divergence saturates for signal and apparatus alike, so only a statistic that nulls under apparatus swaps and moves under model swaps is a model reading).

Two confounds shape every metric. *Repetition:* a lattice looping one corpus n -gram scores high overlap for free, so structure uses a repetition-robust variant (distinct corpus n -grams). *Diversity* (its stability analog): a degenerate low-entropy lattice snaps a perturbation back trivially—low damage

for the wrong reason—so damage is normalized by the model’s own diversity floor (§4.3). We validate on a tiny transformer trained from scratch on tinyshakespeare (census checked against the actual training corpus), then scale across a masked family (bert-tiny through bert-large, $L=2-24$) and two autoregressive families (Pythia-14M-1B [Biderman et al., 2023] and GPT-2 small-xl). A free control: the temperature “phase transition” is, on a finite-size scan, a *crossover*, not a true transition (the order-parameter susceptibility self-averages, $\chi_{\text{peak}} \propto 1/N$), disciplining all “transition” language.

3 Validation: reproducing known metrics

An instrument is only as good as its calibration. Before applying the CA/damage-spreading apparatus to a language model—whose “true” dynamical metrics are unknown—we check it against systems where the answer is known, and we are explicit about which properties each rung does and does not exercise (Fig. 2). The ladder is ordered by how much of the instrument’s actual regime it shares: smooth-limit arithmetic, then *discrete* state with a finite perturbation—the regime the token instrument occupies.

(1) The logistic map: a unit test of the growth-rate arithmetic, not a validation of the instrument. A renormalized tangent estimator—two orbits differing by ϵ , with the separation rescaled to ϵ after every step—recovers the analytic exponent $\lambda(r) = \langle \ln |r(1-2x)| \rangle$ of $x \mapsto rx(1-x)$ across the bifurcation diagram to machine precision at $\epsilon=10^{-9}$. We state plainly why that agreement is not evidence about damage spreading: because the twin is re-anchored to the reference orbit each step, $\log(d/\epsilon) = \log |f'(x)| + O(\epsilon)$, so the estimator is a finite-difference evaluation of the derivative it is compared against. An ϵ -sweep confirms the error is $O(\epsilon)$ (log-log slope 0.79 over six decades; Fig. 1). **A token flip is $O(1)$ in a discrete alphabet—there is no $\epsilon \rightarrow 0$ limit in token space**—so this rung certifies only the smooth limit. Removing the renormalization and using a finite perturbation, which is the instrument’s actual regime, leaves an irreducible bias of ≈ 0.15 nats/step that does *not* vanish as $\epsilon \rightarrow 0$ (Fig. 1, flat curve): the windowed fit on a saturating separation, not the perturbation size, sets the error. The weight-bearing rungs are therefore (3) and (4).

(2) A coupled-map lattice (spatial, still smooth). A ring of diffusively coupled logistic maps recovers the single-site $\ln 2$ at zero coupling, and its coupling dependence is verified against an exact Benettin/Jacobian reference (max discrepancy 1.1×10^{-3}); the exponent is non-monotone in coupling, with a minimum near $\epsilon \approx 0.3$. This estimator also renormalizes, so it inherits the caveat above; it extends the arithmetic to a spatially-extended system but not to discrete state.

(3) Elementary CA criticality classes: a coarse split, decisively. Driving the identical damage-spreading protocol (async updates, CRN twins) with classical elementary CA rules of *known* class (19 rules $\times$ 12 seeds, rule as the unit of analysis) separates the rules whose damage *survives* from those whose damage *dies*. The discriminating quantity is the **ignition probability**—the fraction of lattices in which a single-site flip survives at all—not λ : damage in a discrete CA is bimodal, so a λ averaged over ignited and extinguished runs measures the mixture. Ignition probability is also the physically correct object, being the order parameter of the directed-percolation class to which damage spreading in discrete CA belongs [Grassberger, 1995]. Measured on it, ordered rules sit at 0.046 (bootstrap CI [0.000, 0.102]) against 0.668 (edge) and 0.682 (chaotic): **ordered vs. rest** $p=0.0$, **Cohen’s $d=3.03$** . The finer three-class ordering does *not* survive—edge vs. chaotic gives $p=0.470$ —so we claim only the coarse split. We report this rather than the group-mean λ ordering we first computed, for two reasons: that ordering was not significant edge-vs-chaotic ($p=0.17$ on the same statistic), and its ordered-group mean is not a measurement at all. Five of seven ordered rules return exactly $-0.4 \ln 10$ with a zero-width interval—the value the log-linear fit takes when damage dies immediately and the cone is clamped at the estimator’s numerical floor (DEAD_DAMAGE_FLOOR), identical across rule, seed and lattice. A group mean built from it is arithmetic, not an exponent.

(4) Synthetic Markov sources (known transition matrices). The instrument’s attractor census, run on tiny models trained on three synthetic first-order Markov sources with known sparse matrices P_a, P_b, P_c , recovers each model’s own matrix (row total-variation 0.22) and not the others (cross 0.95; random baseline 0.91)—quantitative recovery of a ground-truth generative rule and clean source discrimination (details in §4.6).

(5) The Domany–Kinzel automaton: an exact check on the damage machinery. The DK model [Domany and Kinzel, 1984] is the only rung that is stochastic *and* discrete—the instrument’s own

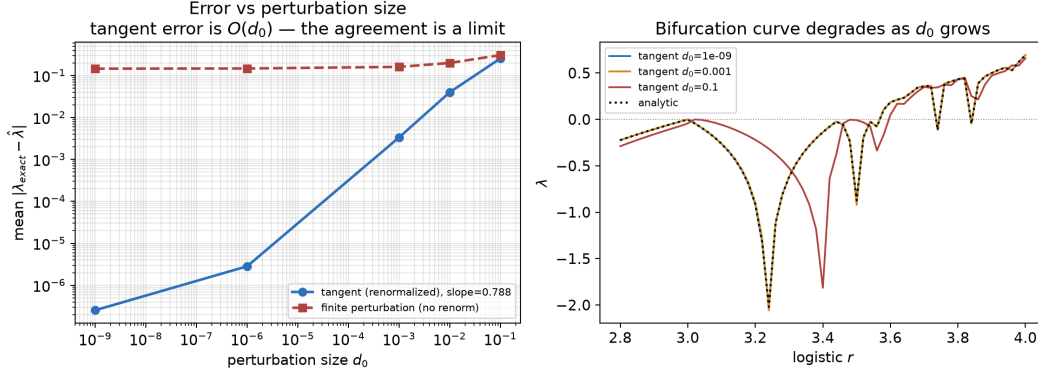


Figure 1: Why the logistic rung is a unit test, not a validation. Left: the renormalized tangent estimator’s error is $O(\epsilon)$ (log–log slope 0.79), so its agreement with the analytic exponent is a property of the infinitesimal limit; the finite, un-renormalized estimator—the instrument’s regime—has an irreducible ≈ 0.15 bias floor that does not improve as $\epsilon \rightarrow 0$. Right: the recovered bifurcation curve degrades visibly once ϵ is not infinitesimal.

regime—and the only one whose damage behaviour has a published boundary rather than an analogy. It also affords something no other rung does. On its $p_2=0$ line the damage field between two commonly-driven replicas is provably *itself* a DK automaton at the same p_1 [Kohring and Schreckenberg, 1992]: with $s'_i = (s_{i-1} \oplus s_{i+1}) \theta(p_1 - z_i)$, sharing z_i gives $d'_i = (d_{i-1} \oplus d_{i+1}) \theta(p_1 - z_i)$. We therefore predict the damage trajectory with an independent simulator and demand *bit-identity*: across $p_1 \in \{0.2, \dots, 1.0\}$, a ring of 4096 over 1500 steps, **zero mismatching cells**, with 16 mismatches in an off-line control at $(0.6, 0.5)$ where the mapping must break. This is run through the same simulation loop that produces every language-model number below, so the window indexing, the shared-uniform consumption order, the inverse-CDF sampling and the update schedule are verified with no error bar attached. A second closed form rides inside the same run: at $p_1=1, p_2=0$ the update is deterministic $\oplus$ —Rule 90—so a single-seed damage field must hold exactly $2^{\text{popcount}(t)}$ live cells; at $t=1500$ that is $128/4096 = 0.03125$, which is the density obtained, and the cone (3001 cells) never reaches the boundary.

The critical points are a weaker check and we label them so: the local-slope estimator ($\delta_{\text{eff}} \rightarrow \delta_{\text{DP}}=0.1595$, 2000 seed runs) gives $p_c=0.7065$ on the site-DP line (0.15% from the published 0.705489(4)) and 0.8092 on the $p_2=0$ line, with the damage field measured independently at 0.8089—coinciding with the activity transition to 3×10^{-4} , as the mapping requires. The published $p_2=0$ value is itself disputed, 0.801(2) [Zebende and Penna, 1994] versus 0.8087(5) [Hinrichsen et al., 1997], and at our stated $\sim 1\%$ accuracy we cannot discriminate between them; we report both and claim neither.

The rungs that share the instrument’s regime—discrete state, finite $O(1)$ perturbation, no renormalization—are (3), (4) and (5), and those are what license its use on language models; (5) is the only rung on which agreement is exact rather than fitted. The smooth-limit rungs are retained as arithmetic unit tests and labelled as such.

4 New measurements on language models

Pointed at a trained LM, the validated instrument yields the quantities below. We are precise about what is new: several of these *quantities* have been identified in concurrent work (a critical sampling temperature [cri, 2026, 2025a,b, Nakaishi et al., 2024, Arnold et al., 2024]; edge-of-chaos and Lyapunov framings [Bertschinger and Natschläger, 2004, Tomihari and Karakida, 2025]). What we contribute is not the concepts but a single, CRN-certified, ground-truth-calibrated *route* to them from weights-free access—the same estimator validated in §3—together with a few apparatus-level observations (the radius law, the confound taxonomy) that are specific to the token-lattice construction.

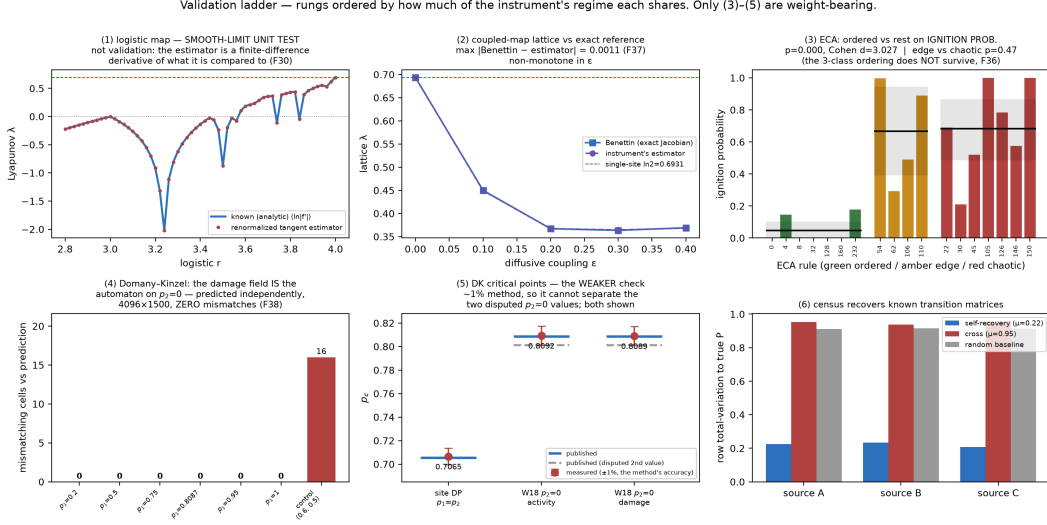


Figure 2: Validation ladder, ordered by how much of the instrument’s regime each rung shares. (1) logistic-map Lyapunov—a *smooth-limit arithmetic unit test* (the tangent estimator is renormalized, so its agreement is an $O(\epsilon)$ limit property, not evidence about damage spreading; see Fig. 1); (2) coupled-map-lattice finite-size Lyapunov ($\ln 2$ uncoupled, known coupling reduction); (3) elementary-CA criticality, where the instrument separates rules whose damage survives from rules whose damage dies (ignition probability, ordered vs. rest $d=3.03$; the finer three-class ordering does *not* survive, $p=0.470$); (4) the census recovering known Markov transition matrices (self-TV ≈ 0.22 vs. cross ≈ 0.95).

4.1 A radius law

Sweeping r at fixed T , the *local* statistic (corpus-bigram fraction) is nearly radius-blind. *Longer-range* corpus consistency is not: raw 3- and 4-gram overlap rises with r , but under the repetition-robust metric the signal *peaks at an intermediate radius* ($r \approx 4$) and falls by $r=16$ —the apparent large- r growth was collapse into repetition. Certification: holding the special-token windowing scheme fixed, the cross-model radius-profile difference far exceeds the distribution-preserving apparatus floor (so the profile is a model effect), but the scheme itself moves the profile as much as a model change, so cross-model claims must hold the scheme fixed (Fig. 3). Sweeping r at fixed T , the *local* statistic (corpus-bigram fraction) is nearly radius-blind. *Longer-range* corpus consistency is not: raw 3- and 4-gram overlap rises with r , but under the repetition-robust metric the signal *peaks at an intermediate radius* ($r \approx 4$) and falls by $r=16$ —the apparent large- r growth was collapse into repetition. Certification: holding the special-token windowing scheme fixed, the cross-model radius-profile difference far exceeds the distribution-preserving apparatus floor (so the profile is a model effect), but the scheme itself moves the profile as much as a model change, so cross-model claims must hold the scheme fixed (Fig. 3).

4.2 Damage light cones: $v \propto r$

A 3-site block flip under CRN produces a cone whose front advances ballistically at $v \approx 1.6, 3.5, 7.7, \dots$ sites/sweep for $r = 1, 2, 4, \dots$. A plateau at large r on small rings is finite-size: the front fills the ring in ~ 2 sweeps, so the velocity measured there is merely the wraparound ceiling $\approx N/4$ (the values 11.5, 23.5, 47.5 for $N=48, 96, 192$ at $r=16$ are exactly $N/4$, not an unclipped law). Only the largest ring escapes clipping at $r=16$ ($N=384$), and there the velocity (41.1) sits *below* the smaller-ring ceilings—confirming they *were* ceilings—while the unclipped points are mildly *super-linear* ($\approx 1.5r$ at $r=4$ rising to $\approx 2.6r$ at $r=16$). So we do not claim a clean $v \propto r$; what is robust and **model-invariant** is that the front velocity grows monotonically with the conditioning radius—interaction range is a kinematic property of the apparatus, not the model (Fig. 4). We do *not* claim even the qualitative law as new: a ballistic damage front whose velocity is set by the neighborhood range is the classic CA light cone—the maximal Lyapunov exponent equals the damage-front

A1: is the radius-profile shift a MODEL effect once the scheme is held fixed?

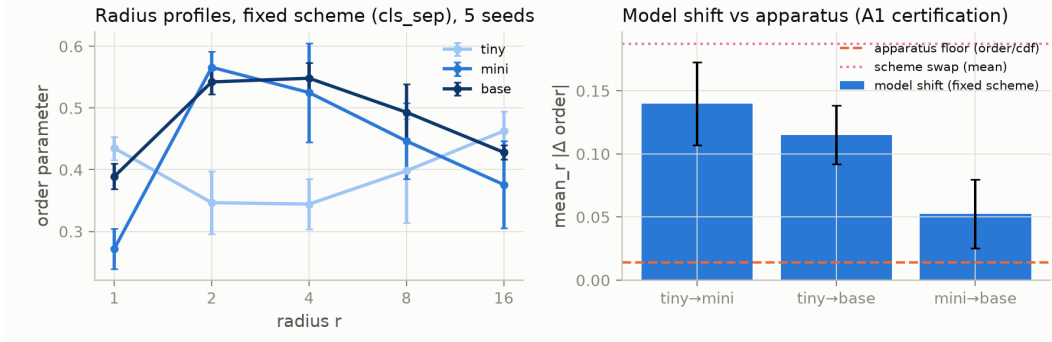


Figure 3: Radius certification. Cross-model radius profiles at a fixed scheme exceed the distribution-preserving apparatus floor (model effect), but the special-token scheme is a first-class apparatus of comparable magnitude.

A2: is the $r=8,16$ velocity plateau a finite-size artifact?

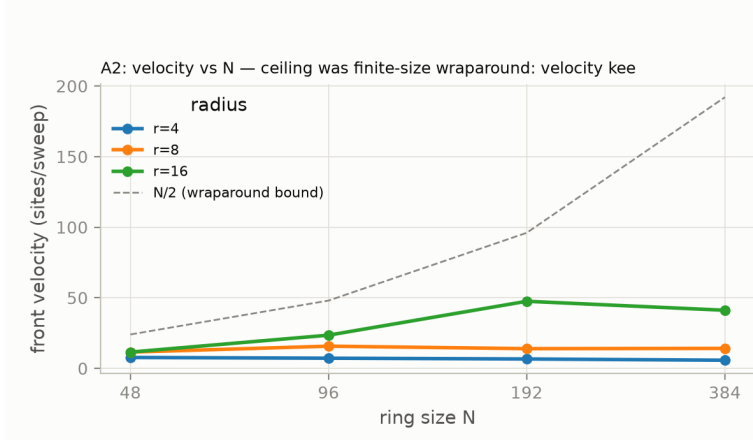


Figure 4: Damage-front velocity vs. ring size. The small-ring plateau is finite-size wraparound; $v \propto r$ continues once unclipped.

203 velocity [Bagnoli et al., 1992], finite-range interactions give a constant range-set velocity by the
 204 Lieb–Robinson bound [Lieb and Robinson, 1972], and butterfly velocities of decorrelator fronts are
 205 standard in chaotic CA [Liu et al., 2021]. Our point is that the LM token-lattice *behaves as a bona-*
 206 *fide CA*—it exhibits this light cone—and that the velocity is model-invariant, i.e. a shared kinematic
 207 axis (contrast §4.3).

208 4.3 The damping length

209 Perturb, evolve under CRN, and ask how much damage *persists*. The raw answer is diversity-
 210 confounded: a low-entropy lattice snaps back for free. We therefore report the perturbation damage
 211 D alongside a **diversity floor** D_0 —the asymptotic drift of twins sharing the settled initial state
 212 but evolving under *independent* noise with *no* flip—and their ratio $D_{\text{norm}} = D/D_0$. Two caveats
 213 on the normalization, stated plainly because they bound the reading. *First*, D and D_0 use different
 214 couplings (D common-random-number, D_0 independent noise), so D_{norm} is a controlled diagnostic,
 215 not a ratio of like quantities; the natural CRN floor is the null itself ($= 0$). *Second*, at large r the
 216 rise in D_{norm} is **denominator-driven**: raw D already *falls* (tiny, $T=0.7$: $0.54 \rightarrow 0.39$ from $r=4$
 217 to $r=16$) while D_0 falls faster, so “the damping length keeps shrinking” at large r partly reflects
 218 the floor collapsing, not a stronger perturbation response. *Third*, and more fundamentally, damage

spreading is a property of (*model, coupling*) rather than of the model alone—a known result in stochastic PCAs, where the damage boundary moves with the replica-coupling algorithm while single-replica dynamics is entirely insensitive to it [Kohring and Schreckenberg, 1992, Grassberger, 1995]. We can say precisely where our choice sits, and we are careful about how far that extends. Drawing one uniform per site and thresholding it against every probability is, on a *binary* alphabet, exactly the *maximal-correlation* member of the admissible family in the classification of Hinrichsen et al. [1997]—which is why the Domany–Kinzel identity above is exact. On a vocabulary, it is not: inverse-CDF sampling from a shared uniform is the *monotone* coupling, and for $|V| > 2$ the two differ (for $p=(\frac{1}{2}, \frac{1}{2}, 0)$, $q=(0, \frac{1}{2}, \frac{1}{2})$ the maximal coupling agrees with probability $\frac{1}{2}$ and the monotone one never). Since maximal coupling maximises agreement and hence minimises damage, our language-model damage numbers are *not* an extremum of the family; they sit inside it. Measured on real conditionals from a live damage run, the excess disagreement our coupling introduces is small—1.3% ($T=0.7$) to 5.4% ($T=0.9$) above the maximal-coupling value on average, and a factor 1.2–1.4 in the near-agreement regime ahead of the damage front. The property that justifies the choice is a different one: inverse-CDF is **replica-independent**—each replica’s next state is a function of its own state and the shared noise alone, never of its twin—so it extends consistently to more than two replicas and to a self-consistent damage field, whereas a maximal coupling is defined only pairwise. Relative comparisons (across checkpoints, across radii, rule to rule) are unaffected, since the coupling is a common mode; the absolute scale of D_{norm} carries this caveat.

Read as a floor-relative diagnostic (Fig. 5), D_{norm} rises with radius: the token-space damping length shortens as the window widens. This is the measurement we report—a diversity-controlled, weights-free error-persistence length **of the token-lattice construction**. That attribution is deliberate and is the subject of §4.4: it is *not* a property of free generation. A capacity-linked gap does appear among masked models (mean D_{norm} tiny $0.672 < \text{mini } 0.880 \approx \text{base } 0.904$), but we *do not build on it*: it rests on two seeds and one toy family, the only clean separation is tiny $\ll \{\text{mini, base}\}$, and an earlier “ $p < 10^{-4}$ ” statistic for it was pseudoreplicated (retracted—see Limitations and the repository audit). We likewise withdraw a previously-reported kinematics-versus-stability decomposition, which held only at the single grid cell where the early Lyapunov exponent saturates. Whether capacity sets a stability↔expressiveness axis [Bertschinger and Natschläger, 2004, Zhang et al., 2025] is left open, pending a seed-level rebuild; the damping length itself stands as a measurement independent of that question.

4.4 What the damping length measures: free generation absorbs nothing

The instrument’s central assumption—that a number measured on the ring says something about the model—is one we tested rather than assumed, by running the identical damage protocol on *real autoregressive generation*. From a prompt we generate two CRN twins sharing one uniform stream, force a different token at one position in one twin, and let both continue freely. The null arm (no injection) is asserted to diverge by exactly zero, and does.

The result is unambiguous and negative: on pythia-70m, -160m and -410m (32 trials each), the injected error **persists in every single trial** ($P_{\text{persist}}=1.000$, $P_{\text{reconverge}}=0.000$, mean divergence fraction 0.94–0.96). Token identity is a harsh readout after an injection shifts alignment, so we also measured distributionally: the total-variation distance between the twins’ next-token distributions, normalized by the TV between two *independent* continuations of the same prompt, settles at $TV_{\text{norm}} \approx 0.97$ —the twins end as far apart as two unrelated continuations.

The mechanism is structural, not a property of any particular model. **Free generation never resamples a token**: once emitted, an error stays in context permanently. The ring CA revisits every site, and in-place resampling is precisely what creates the possibility of healing. So the damping length, D_{norm} and the light cone characterise *the iterated resampling construction*—they are not predictions about autoregressive decoding, and we do not offer them as such. This retroactively explains three otherwise-puzzling results: the light-cone velocity that depends on r but not the model, the model-invariance of $\lambda_{\text{ca}}(r)$, and the structural failure of the cross-level proxy (§5).

What this does and does not cost the paper. It removes any reading of these quantities as error-correction in deployed decoding. It does *not* touch the developmental result, and the reason is worth stating explicitly: the construction is held *fixed* across checkpoints—same ring, same radius, same temperature, same coupling, same seeds—so a change measured across checkpoints cannot come

F23: diversity-controlled damage D_{norm} rises with r and separates capacity (tiny $\ll$ {mini, base}); norm spread=0.232)

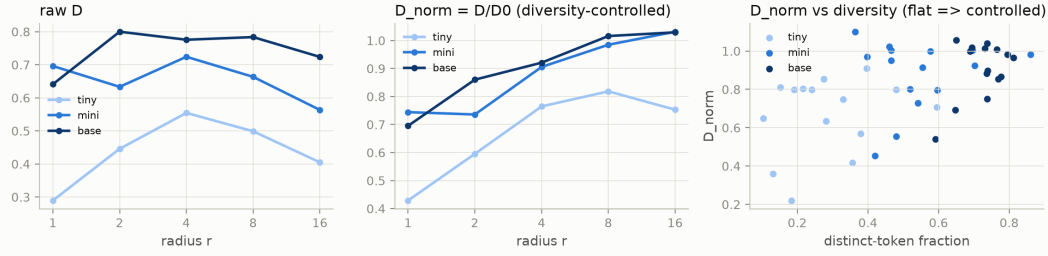


Figure 5: The damping length. Left: raw damage D vs. r . Middle: diversity+velocity controlled $D_{\text{norm}} = D/D_0$ —rises with r ; $D_{\text{norm}} \gtrsim 1$ for the larger models. Right: D_{norm} vs. diversity (flat $\Rightarrow$ controlled).

from the apparatus. It is attributable to the model. A fixed instrument giving different readings on different models is exactly what an instrument is for; what F35 forbids is transporting the reading’s units to a process the instrument does not implement.

4.5 External validity: the autoregressive port

Both load-bearing measurements replicate on Pythia-160m driven through a one-sided *causal* window (null CRN divergence exactly zero). Velocity $\propto r$ holds ($v = 5.8, 7.7, 11.5$ for $r = 2, 4, 8$; $r=1$ does not propagate), and the damping length holds (D_{norm} climbs $0.001 \rightarrow 0.98$ across r). The capacity $\rightarrow$ sensitivity result, however, does **not** robustly carry over. Across four Pythia sizes with 5 seeds (mean D_{norm} at $r=2$, $\pm$ SE): 0.41 ± 0.08 (70m) $\approx 0.41 \pm 0.08$ (160m) $< 0.56 \pm 0.06$ (410m) $> 0.47 \pm 0.08$ (1b)—non-monotone, peaking at 410m and dropping at 1b, with an insignificant size-rank correlation (Spearman $\rho=0.17$, $p=0.29$; only 160m $\rightarrow$ 410m is marginal, $p=0.06$). So the masked-side capacity ordering (§4.3) is masked-specific in this data, not construction-independent; neither construction extends it monotonically (masked saturates, AR reverses). What replicates on AR is the *instrument* (velocity grows with r ; the damping length rises with r), not the capacity scaling. The masked-LM local conditionals are globally inconsistent [Sana et al., 2026]; the autoregressive rule here is a *consistent* order- r causal kernel—but resampled in place on a periodic ring, so it too is not a sampler of the model’s true left-to-right joint. Both constructions are thus windowed, in-place-resampled ring dynamics rather than the model’s actual distribution. Replication across them is evidence that the radius-level measurements are not tied to the masked windowing—not proof about either joint. We compare at the level of the *trend*, since causal-context healing is a different object from bidirectional healing. (The AR means pool two temperature regimes with very different D_{norm} scales; per- T reporting is the correct fix, flagged in Limitations.)

Crossover-relative robustness (is the plateau a sampling artifact?). A fixed (r, T) probe could sample different points of each model’s phase diagram, so we locate each model’s heal $\rightarrow$ spread crossover by sweeping T at $r=2$ (Fig. 7). *Masked*: base’s profile is clearly shifted up (it amplifies at lower T) and tiny is lowest across most of the range, consistent with the tiny $\ll$ {mini, base} gap; but the ordering is *not* clean at every T (at $T=0.3$, mini $0.46 <$ tiny 0.51) and the profiles are single-seed with no error bars. We therefore read this as a plateau *diagnostic*—the mid-vs-large saturation is a large-radius averaging effect—rather than an independent confirmation of a monotone capacity axis. *Autoregressive*: the ordering does *not* hold—Pythia-410m is an outlier (profile shifted up, lowest crossover) while $70\text{m} \approx 160\text{m} \approx 1\text{b}$ —so the AR non-monotonicity is genuine, not a sampling artifact. (The extracted crossover temperature is itself noisy in the high-variance low- T healing regime—bert-tiny’s profile is non-monotone there—so we compare profiles at fixed T rather than lean on it.)

4.6 The attractor census

The census is the instrument’s read-out of internalized distributional structure; its ground-truth validation (rung 4 of §3) is here in full. On three synthetic first-order Markov sources sharing a vo-

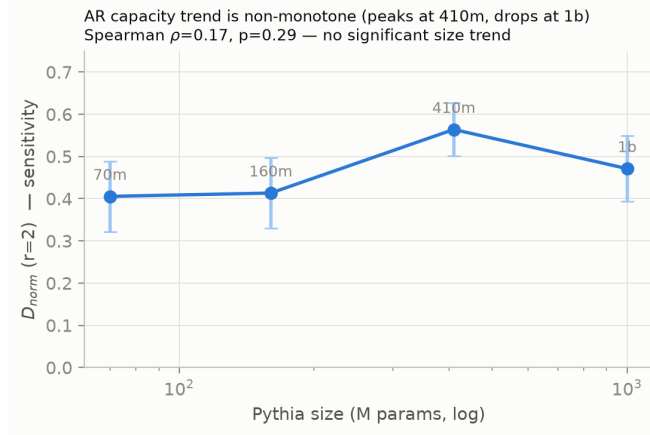


Figure 6: The AR capacity trend does not hold. Diversity-controlled D_{norm} at $r=2$ vs. Pythia size (5 seeds, $\pm$ SE): non-monotone, peaking at 410m and dropping at 1b; Spearman $\rho=0.17$, $p=0.29$. The masked-side capacity effect is masked-specific in this data.

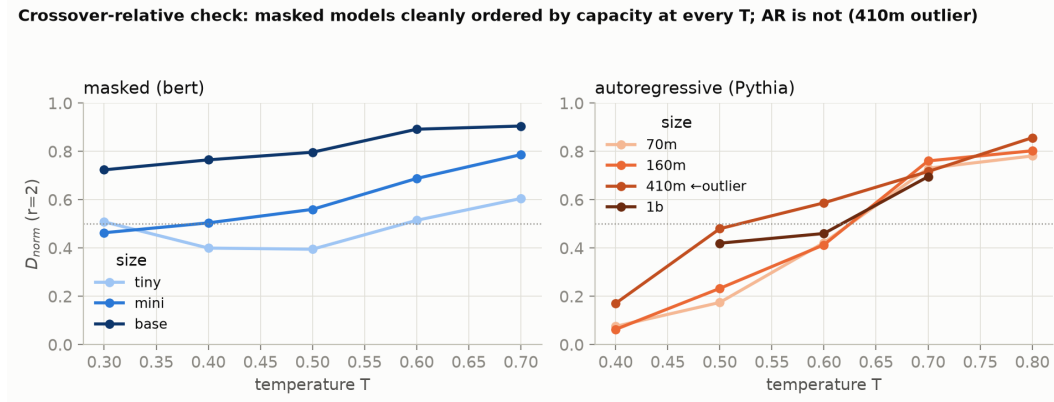


Figure 7: Crossover-relative check: $D_{\text{norm}}(T)$ at $r=2$ (single-seed). Masked models (left): base is shifted up and tiny lowest across most of T (the mid-vs-large plateau is a large-radius averaging effect), though the ordering is not clean at every T ; the AR models (right) are not ordered—Pythia-410m is an outlier and the others overlap, so the AR non-replication is real.

cabulary but with different sparse transition matrices $P_a/P_b/P_c$, a model trained on each and censused yields an empirical bigram transition Q_X matching its own P_X (row total-variation 0.22 vs. a random-lattice baseline 0.91) and *not* the others (cross-TV 0.95): the census recovers each model’s own priors quantitatively and discriminates sources (Fig. 8)—the honest trained-model analog of a sampler oracle [Tang et al., 2026]. This quantitative recovery is established *only* on the synthetic toy trained on the exact sources. On real models the census is scored against an out-of-training WikiText proxy and the numbers are near-floor (top-50 overlap 0.02–0.04, Spearman 0.14–0.21); the largest model’s deepest attractors resemble a corpus *format skeleton*, but the smallest model’s are degenerate token fragments. So the “the census reads out what the model internalized” reading is licensed by the synthetic calibration, not by the real-model overlaps. Vocabulary compression corrupts even that—a word-level $\langle \text{unk} \rangle$ becomes attractor material and inflates recovery through artifact-to-artifact matching; byte-level BPE removes it.

5 The instrument’s boundary: token- vs. activation-space criticality

A black-box instrument earns its keep if its readings track something one would otherwise need internals to see. We tested this hypothesis thoroughly—a ground-truth calibration, two model families,

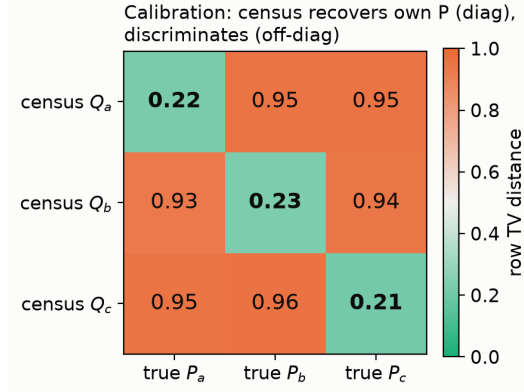


Figure 8: Calibration. Row total-variation between each model’s censused bigram transition Q_X and every true P_Y : diagonal (self-recovery) ≈ 0.22 , off-diagonal ≈ 0.95 —quantitative recovery and clean discrimination.

two within-model axes—and report a clean **negative**: the token-space instrument does *not* robustly proxy activation-space criticality.

The instrument separates ordered from chaotic rules. Driving the same damage-spreading estimator (async, CRN, single-site flip) with 19 classical Elementary CA rules of *known* class over 12 seeds, the rule-level bootstrap separates ordered from the rest decisively ($p=0.0$, Cohen’s $d=3.03$), while edge-vs-chaotic does not separate ($p=0.470$); we therefore claim only the coarse split. The discriminating quantity is the *ignition probability* (ordered 0.046; edge 0.668; chaotic 0.682), because single-site damage is bimodal—it either ignites or dies—so a λ averaged over both populations measures the mixture rather than a growth rate. This is the criticality-side analog of the census calibration, at the resolution the data support.

The white-box reference is a clean, cross-family scaling law. On the autoregressive ladder we compute a finite-depth top-Lyapunov exponent $\lambda_{\text{top}} = \frac{1}{L} \log \rho(J_{\text{emb} \rightarrow h_L})$ (the SPARC error-propagation operator $\rho(F_T)$ [Petrova and Vejsiu, 2026]; finite-difference-JVP power iteration, fp32). It falls monotonically with size (Pythia 14M–1B, Spearman $\rho = -0.94$, $p=0.005$) and this replicates across families (GPT-2 124M–1.5B, $0.152 \rightarrow 0.059$): larger models sit nearer critical from the supercritical side.

But the black-box instrument does not proxy it (Fig. 9). (i) *Cross-model*, fixed (r, T) : the type-matched pairing (λ_{top} vs λ_{ca}) is suggestive on Pythia ($r=0.71$, $n=6$) but does *not* replicate on GPT-2 ($r = -0.43$); pooling the families gives $r=0.70$, $p=0.025$, but that is a Simpson’s-paradox artifact of two disagreeing families, not a real effect (the persistence quantity D_{norm} does not track either, $r = -0.29$). (ii) *Within-model across temperature*: $\lambda_{\text{ca}}(T)$ vs $\rho(F_T)$ gives a uniform $r \approx -0.9$ for all six models and both families—a mechanical confound (raising T makes the CA explore more while flattening the softmax), betrayed by its very uniformity, not cross-level signal. (iii) *Within-model across radius* (the de-confounded axis, fixed T): the correlation is null (per-model r centred on 0), because $\lambda_{\text{ca}}(r)$ is itself *model-invariant*— $\approx [-1.1, 0.15, 0.5, 0.9]$ for $r=1, 2, 4, 8$ across every model and family, pure light-cone kinematics carrying no model signal, while $\rho(F_r)$ varies by model.

Conclusion: token-space and activation-space criticality are **distinct, non-proxying levels**. The instrument measures a real, calibrated token-space criticality, but it is not a weights-free surrogate for the activation-space Jacobian—the natural proxy hypothesis is false, cleanly. The instrument’s value is therefore as a *token-space* measurement in its own right, not as a stand-in for internals.

The negative is structural, not a measurement failure. We mined for a positive along the principled axes and hit the same barrier each time: λ_{top} is an *architectural* quantity. Across ten Pythia-410M training checkpoints it is nearly flat (~ 0.13 – 0.19) while the black-box criticality undergoes a clean order→chaos transition around step 10^3 —the levels do not co-evolve. On a six-model BERT depth ladder ($L=2$ – 24) the white-black correlation is strong ($r = -0.92$) but purely *depth-mediated*: both track depth ($r = -0.92$ and $+0.76$), and at fixed depth (two $L=4$ models) λ_{top} is identical while

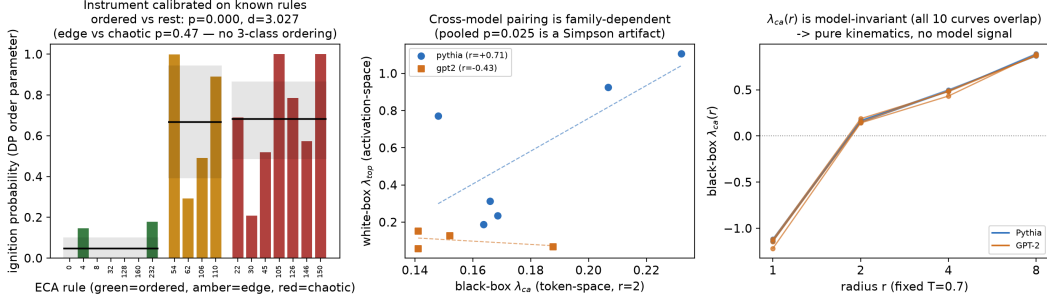


Figure 9: Cross-level. Left: the instrument is criticality-calibrated—it separates classical CA rules whose damage survives from those whose damage dies (ignition probability, ordered vs. rest $p=0.0$, $d=3.03$; bars are per-rule, black lines group means, shading the bootstrap CI). Middle: the cross-model pairing is family-dependent (Pythia $r=0.71$ vs. GPT-2 $r=-0.43$; the pooled $p=0.025$ is a Simpson’s-paradox artifact). Right: black-box $\lambda_{ca}(r)$ is model-invariant (all six models, both families, overlap)—pure kinematics, so it carries no model signal to proxy the white-box $\rho(F_r)$.

363 D_{norm} differs—so the correlation conveys only the (public) layer count. Because $\lambda_{\text{top}} \approx 1/L$ is set
 364 by architecture rather than learning, no *useful* weights-free proxy exists with these measures; the
 365 decoupling is a structural fact about the two levels, not an artifact of too little data.

366 6 Related work

367 A novelty check (102-agent adversarial harness) places our contribution precisely: the *instrument*
 368 is largely novel at the method level, but its *substrate* is not, and novelty is uneven across claims.
 369 **Shared substrate (the exposed flank).** Sana et al. [2026] already recasts a masked LM as iterated
 370 masked-token resampling—a Glauber Markov chain on token sequences—the exact substrate we
 371 iterate. It measures mixing time ($O(n \log n)$ high- T) and low- T metastability, not a conditioning-
 372 radius axis, light-cone velocity, a damping length, an attractor census, or a finite-size Lyapunov
 373 exponent. The distinction is sharp in the coupling: it tracks Hamming-distance *contraction* under
 374 *maximal* coupling to *bound mixing*, whereas we track Hamming-distance *spatial propagation* under
 375 *common-random-number* coupling to measure the light cone and a diversity-normalized damping
 376 length across a swept radius—maximal coupling minimizes disagreement (a mixing device), CRN
 377 preserves the common noise so the perturbation’s causal spread is visible. Our novelty rests on this
 378 measurement layer, not on the LM-as-iterated-token-CA framing, which we do not claim. **Edge-**
 379 **of-chaos (claim of §4.3 is partially anticipated).** Measuring proximity to the edge of chaos and
 380 tying it to capability is canonical [Langton, 1990, Bertschinger and Natschlger, 2004] and has
 381 been shown for trained transformers via Lyapunov exponents of self-attention Jacobians [Tomihari
 382 and Karakida, 2025], and named for LLMs generally [Zhang et al., 2025, Li et al., 2025]. We
 383 claim neither the concept nor the criticality↔capability link; our contribution is the *token-space*,
 384 black-box damping-length measurement and its certification/calibration protocol, which Tomihari
 385 and Karakida [2025] (white-box, continuous hidden space) does not provide. We do *not* claim an
 386 established model-size→sensitivity axis (see §4.3). **The CA light cone is classic (§3.2 is import,**
 387 **not discovery).** A ballistic damage front whose velocity is set by the neighborhood range, and the
 388 identity Lyapunov=front-velocity, are textbook CA / statistical-physics [Bagnoli et al., 1992, Lieb
 389 and Robinson, 1972, Liu et al., 2021]; we import them and show the LM token-lattice exhibits them
 390 (behaves as a bona-fide CA) and that the velocity is model-invariant—not that the law is new. **Per-**
 391 **turbation propagation.** Petrova and Vejsiu [2026] formalizes an error-propagation operator on AR
 392 residual streams with a $\rho(F_T) \geq 1$ criticality threshold (the top-Lyapunov boundary) in activation
 393 space, adjacent but not a token-space CA; Nilayam and Nayak [2026] operates on compound-AI
 394 pipeline graphs, sharing vocabulary but no method. **Terminology.** We avoid “repair”—taken by
 395 self-repair / the Hydra effect [McGrath et al., 2023, Rushing and Nanda, 2024] for internal compo-
 396 nent compensation—and “self-correction” [Petrova and Vejsiu, 2026], both categorically different
 397 from our spatial damping length. **Critical sampling temperature (the quantity, not the method,**
 398 **is prior work).** A per-model critical temperature separating ordered from disordered generation is
 399 already defined and measured across at least five independent works: cri [2026, 2025a,b] locate a T_c

by sweeping the softmax temperature (susceptibility peak, order-parameter collapse, or a measured critical region $T \approx 1.2\text{--}1.4$), and Nakaishi et al. [2024], Arnold et al. [2024] identify the same temperature-driven order→disorder transition via power-law statistics and phase mapping. The damage-spreading-to-Lyapunov estimator we use is likewise classical CA physics [Bagnoli et al., 1992]. We therefore *do not* claim an edge-of-chaos temperature or a token-space Lyapunov as new quantities; our contribution is a CRN-certified, ground-truth-calibrated damage-spreading *route* to them (a different estimator: Lyapunov zero-crossing rather than susceptibility). **Activation-space propagation.** tft [2026] builds an empirical finite- ϵ position×depth residual-stream propagation cone and res [2026] its analytic Jacobian counterpart; these anticipate the activation-space cone we sketch as future work (§8), so we treat that direction as a rigor/calibration extension, not a new object. **Other.** Tang et al. [2026] is the calibration/oracle kin (our synthetic-Markov recovery is its trained-model analog); the census descends from Hanson and Crutchfield [1997]; early low-order crystallization is consistent with distributional simplicity bias [Refinetti et al., 2023], read black-box. (*Pre-submission: full-paper reads of Sana et al. [2026], cri [2026], tft [2026] and Nilayam and Nayak [2026]; verify all 2026-preprint titles/IDs; see results/deep_research_novelty*.md.*)

7 Limitations

Masked-LM local conditionals are globally inconsistent, so the CA is a well-defined stochastic dynamical system but not a sampler of any joint; all claims are properties of the dynamics. The real-model census is proxy-validated (WikiText, out-of-training), a lower bound; ground-truth calibration is on synthetic sources only. Rings are small ($N \leq 384$), the ladder short (three masked, two autoregressive), and the AR capacity trend rests on two models with only the sub-saturation radius discriminating (the 410m largest window OOM’d on 16 GB). The special-token windowing scheme is a first-class apparatus factor to hold fixed; temperature scales are not comparable across masked and autoregressive constructions.

Statistical scope (what the numbers do and do not license). The paper’s central claims—the validation-ladder reproductions (§3)—are exact or near-exact matches to known values and do not rest on null-hypothesis testing, so multiple comparisons are not a concern for them. Where we *do* report trend tests (the cross-level correlations, §5), we apply no formal multiplicity correction and treat them as exploratory: the reported conclusion is a *negative* reached by sign-consistency across families and axes, not by any single surviving p -value, so it is robust to the correction we omit. The one demoted capacity gap is measured at $N=48$ with two seeds and is not built upon; establishing it as an *axis* would need ≥ 15 independent seeds tested at the seed level (hierarchical / bootstrap, not a signed-rank over correlated cells), a longer second-family ladder, multiple $N \gg 100$, per- T (not pooled) reporting, and the diversity-floor normalization ablated against the CRN-null and maximal-coupling floors—all enumerated in the repository tracker as optional pre-camera-ready work, not load-bearing for the validated-instrument contribution.

8 Conclusion

The contribution of this paper is a *validated* black-box instrument. Its organizing principle—validation by reproduction—is what separates it from the qualitative iterated-LLM literature: before measuring a language model, the CA/damage-spreading apparatus separates elementary CA rules by whether damage survives, reproduces a stochastic PCA’s published damage-spreading boundary—bit-exactly on the line where an exact mapping exists—and recovers the known transition matrices of synthetic Markov sources: the rungs that share its regime of discrete state and finite perturbation, while the smooth-limit logistic and coupled-map rungs are retained, and labelled, as arithmetic unit tests whose agreement is an $O(\epsilon)$ limit property (§3). An instrument that recovers what is independently known, and that is explicit about which of its checks are load-bearing, can be trusted on what is not. Pointed at trained LMs, it then yields weights-free measurements a next-token evaluation does not (§4): a token-space Lyapunov exponent and a diversity-controlled damping length of the lattice construction, an effective interaction radius ($r \approx 4$), a model-invariant damage light cone, and an attractor census of internalized structure—each certified by the exact-zero-null CRN protocol and, where ground truth exists, calibrated against it. We also draw the instrument’s *boundary* (§5): a controlled cross-level study returns a clean negative—the black-box token-space criticality does not proxy the white-box activation-space Lyapunov, because the latter is architectural—so token- and

activation-space criticality are distinct, non-proxying levels. The phenomena the instrument measures are old and deep; what is new is a trustworthy way to measure them from the outside, and an honest map of where that measurement reaches its limit.

Opportunities and future work. The instrument reads a model’s dynamics from *token space alone*—no weights, gradients, or activations—which makes it usable in the black-box / API-only regime. *Applications.* (i) The damping length, light cone, and critical sampling temperature [cri, 2026] are measurable on closed models where activation-space probes are unavailable; our cross-level study (§5) maps precisely how far this black-box reading can and cannot substitute for internals. (ii) The attractor census, calibrated to recover a *known* transition matrix (§4.6), is a quantitative probe of the distributional structure a model has internalized [Hanson and Crutchfield, 1997]. (iii) The effective interaction radius ($r \approx 4$) can be correlated with attention-head span or induction heads to link a black-box observable to mechanism. (iv) CRN damage cones are a causal-influence map over token positions. We deliberately do *not* offer the damping length as a bound on error propagation in deployed decoding: §4.4 shows free generation absorbs no injected token error at all. (v) The training-time trajectory (F25) is a phase-change detector across checkpoints for developmental interpretability. *Extensions of the same validated method.* (vi) A second, *compositional-complexity* axis—an effective-rule-complexity or assembly-theoretic read-out of what the dynamics *produce*, complementary to the criticality of how they *behave*—would be validated the same way, against strings of known complexity. (vii) Applying the damage-spreading machinery to the residual stream gives an activation-space information-propagation cone; that object is closely anticipated [tft, 2026, res, 2026], so the contribution there would be the CRN certification and ground-truth calibration, not the cone itself. (viii) The deterministic (greedy, $T \rightarrow 0$) limit is a distinct dynamical object worth measuring, to test whether the criticality readings are a sampling phenomenon. *Rigor.* The census recovery, quantitative on synthetic sources, should be re-established against a real model’s *true* training corpus rather than a proxy; and the demoted capacity gap, if revisited, needs the seed-level rebuild set out in Limitations. Throughout, the reusable asset is the statistics-level, apparatus-nulled certification protocol, which applies to *any* iterated-LLM system—self-refinement, agentic loops, iterated rewriting—where trajectory-level divergence otherwise saturates for signal and apparatus alike. A Lieb–Robinson-style propagation bound [Lieb and Robinson, 1972] for the non-unitary stochastic LM dynamics would put the light cone on rigorous footing.

A Reproducibility

All code, checkpoints, per-run results, and figures are in the `token-lattice-ca` repository; each headline maps to a finding Fk in `findings.md`. Hardware: Apple M1 Pro, 16 GB; toy model on CPU JAX, real models fp16 on the MPS backend.

Models. *Toy:* a 0.42 M-parameter bidirectional transformer ($d=96$, 4 heads, 2 layers) trained from scratch on tinyshakespeare (word-level 2000 vocab, and a 4096 byte-level BPE) as a joint windowed-conditional model over $r \in \{1, 2, 4, 8, 16\}$, 6000 Adam steps. *Masked:* prajjwal1/bert-tiny (2L/128), -mini (4L/256), bert-base-uncased (12L/768), shared bert-base-uncased WordPiece vocab (30522); loaded as BertForMaskedLM. *Autoregressive:* EleutherAI/ pythia-70m/160m/410m, causal window $p(x_i | x_{i-r..i-1})$.

Rule / automaton. Async random-order single-site Glauber on a ring of N cells; inverse-CDF sampling from the tempered, special-token-forbidden center (masked) or next-token (causal) distribution, computed on-device. CRN twins share init, visitation order, and the uniform stream; the two twins are run as one $2B$ batch with tiled uniforms (bit-identical to two runs). Null test: exactly zero divergence (tests/).

Grids and seeds. Phase sweep: $N=48$, $B=16$, 40 sweeps, $T \in \{0.5, 0.8, 1.0, 1.3, 1.6, 2.0, 2.5\}$, $r \in \{1, 2, 4, 8, 16\}$; multi-seed sweep uses 5 seeds. Radius certification (F20/F22): $N=48$, $B=16$, 5 seeds, $T=0.7$, schemes `cls_sep` and `no-special`. Velocity FSS (F21): $N \in \{48, 96, 192, 384\}$, $r \in \{4, 8, 16\}$, $B=24$. Damping length (F23): $N=48$, $B=32$, settle 12, 25–30 sweeps, 2 seeds, $T \in \{0.5, 0.7, 0.9\}$, $r \in \{1, 2, 4, 8, 16\}$, tail 8; the diversity floor D_0 is the same-init, independent-noise, no-flip drift (a full radius- r run, hence velocity-matched). Finite-size Lyapunov: least-squares slope of $\log(\text{damaged sites})$ over the ballistic window. Calibration (§3.5): three first-order Markov sources, $K=60$ symbols, 6 successors, 300 k tokens each, a tiny model per source (4000 steps); re-

covery is row total-variation between the censused and true transition matrix. Census on real models is proxy-validated against a WikiText-103 slice re-tokenized per model ($\sim 240\text{--}250$ k tokens).

Statistics. F12 (crossover), F21 (velocity), and the F23 radius trend are finite-size scanned. The F23 capacity gap is measured with *two* seeds; the 15/15-cell signed-rank we originally reported ($p < 10^{-4}$) is pseudoreplicated (correlated (r, T) cells driven by the same two seeds) and is *not* treated as a significance claim—the defensible statement is the tiny $\ll \{\text{mini, base}\}$ gap (base-vs-mini null, 8/15). A seed-level rebuild (≥ 15 independent seeds, hierarchical / bootstrap test) is the tracked pre-camera-ready fix.

Known limits of this run. The AR capacity trend uses two–three Pythia sizes with one discriminating radius; the 410m largest-window cell OOM’d on 16 GB and was dropped; the real-model census is a proxy; reference metadata in `refs.bib` is web-verified but 2026 preprints may change.

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NeurIPS Paper Checklist

1. Claims

Question: Do the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract states the contribution as a *validated* black-box instrument: it reproduces known dynamical metrics (the exact logistic-map Lyapunov, coupled-map-lattice and elementary-CA criticality, known Markov transition matrices) before measuring LMs, and the criticality/edge-of-chaos phenomena it lands on are framed as decades-old (measured, not discovered) and cited to concurrent work that defines them (a critical sampling temperature; residual-stream propagation cones). The scope is precise: the LM measurements are a certified, calibrated *route* to already-identified quantities, and the one cross-level proxy hypothesis is reported as a clean *negative*. No capacity scaling law is claimed—the earlier suggestive two-seed gap is disclosed and explicitly not built upon.

2. Limitations

Question: Does the paper discuss the limitations of the work?

Answer: [Yes]

Justification: The Limitations section states: globally-inconsistent MLM conditionals (a well-defined dynamical system but not a sampler of any joint); small models ($\leq 110M$ masked, $\leq 1B$ AR) and small rings ($N \leq 384$); proxy census (WikiText, not the models’ training data); the AR capacity trend rests on few models; the special-token scheme is a first-class apparatus; temperature scales are not comparable across constructions.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [NA]

Justification: The paper contains no formal theorems; it is empirical/measurement work. Connections to the Lieb–Robinson bound and the Bagnoli–Rechtman–Ruffo Lyapunov–velocity relation are cited, not re-derived.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: All code, model checkpoints (toy), per-run result JSON/NPZ, and figures are in the public repository; the Reproducibility appendix gives models, grids, seeds, batch/ring sizes, the diversity-floor and CRN definitions, and the statistical tests. A harness regression test asserts the CRN null coupling is exactly zero.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions?

Answer: [Yes]

Justification: Repository with a README giving per-phase reproduction commands and M1 runtime tables; pinned dependency versions; scripts are idempotent and resumable. [TODO: make the repository public and add the URL at camera-ready.]

6. Experimental setting/details

Question: Does the paper specify all the training and test details?

Answer: [Yes]

Justification: Reproducibility appendix specifies optimizer/steps for the toy model, the exact $(r, T, N, B, \text{sweeps, seeds})$ grids per experiment, the reference corpora, and the metric definitions.

7. Experiment statistical significance

Question: Does the paper report error bars / statistical significance?

Answer: [Yes]

Justification: The paper’s central claims are the validation-ladder reproductions—exact or near-exact matches to known values (the Domany–Kinzel damage identity, which is verified *bit-exactly*—zero mismatching cells—and its critical points to 0.06–0.15%; logistic-map Lyapunov mean error $< 10^{-3}$; census self-TV 0.22 vs cross 0.95)—which are not

null-hypothesis tests, so multiple comparisons do not apply to them. Seed error bars are reported where relevant. Where we *do* test—the ECA class separation and the developmental transition—the unit of analysis is the rule or the run (never the seed-cell), the tests are bootstrap or rank-based, and the developmental family is pre-registered and corrected with Benjamini–Hochberg FDR. The finer three-class ECA ordering was tested, failed ($p=0.470$), and is reported as not surviving. The cross-level conclusion is a *negative* reached by sign-consistency across two model families and two within-model axes, not a surviving p -value, so no multiplicity correction is load-bearing. The demoted capacity gap is disclosed as a suggestive two-seed effect and not built upon; an optional seed-level rebuild is tracked in the repository (issues #1–#2, #10).

8. Experiments compute resources

Question: Does the paper provide sufficient information on compute resources?

Answer: [Yes]

Justification: All experiments run on a single Apple M1 Pro (16 GB); per-phase wall-clock runtimes are tabulated. The toy model uses CPU JAX; real models use fp16 on the MPS backend. Total compute is a few laptop-days.

9. Code of ethics

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: Measurement/interpretability work on public pretrained models and synthetic data; no human subjects, no sensitive data, no new capability that raises misuse concerns.

10. Broader impacts

Question: Does the paper discuss potential positive and negative societal impacts?

Answer: [Yes]

Justification: The instrument is a black-box interpretability tool (positive: measurement of trained-LM token dynamics without weight access, applicable to closed models). It creates no new generative capability; negative societal impact is minimal and indirect. [TODO: one broader-impacts paragraph.]

11. Safeguards

Question: Does the paper describe safeguards for responsible release of high-risk assets?

Answer: [NA]

Justification: No high-risk models or datasets are released; only measurement code and analysis results on public models.

12. Licenses for existing assets

Question: Are existing assets (code, data, models) properly credited and licenses respected?

Answer: [Yes]

Justification: Pretrained models (BERT, Pythia), datasets (tinyshakespeare, WikiText-103), and libraries (JAX, PyTorch/Transformers) are cited; usage is within their licenses. [TODO: list exact licenses in the appendix at camera-ready.]

13. New assets

Question: Are new assets introduced in the paper well documented?

Answer: [Yes]

Justification: The instrument (code), the synthetic-Markov calibration corpora, and the toy checkpoints are documented in the repository README and the Reproducibility appendix.

14. Crowdsourcing and research with human subjects

Question: N/A.

Answer: [NA]

Justification: No human subjects or crowdsourcing.

15. Institutional review board (IRB) approvals

Answer: [NA]

Justification: No human-subjects research.

16. Declaration of LLM usage

Question: Does the paper describe LLM usage if it was an important, original, or non-standard component?

Answer: [Yes]

702 Justification: Pretrained LLMs are the object of study (the CA rule), documented through-
703 out. [TODO: disclose any LLM assistance in writing/experimentation per venue policy.]